

Nuclear-model comparison against MicroBooNE charged-current data:

$$\text{N/LFG} + F_A^{\text{LQCD}} + \text{hA2018}$$

Summary

Figure 1 compares the GENIE prediction built from a nucleon-level local Fermi gas (N/LFG) nuclear ground state, a lattice-QCD axial form factor (F_A^{LQCD}), and the hA2018 final-state-interaction (FSI) cascade —sample `LFG26_24a_00_000` (solid)— against the MicroBooNE charged-current cross-section data (points) and two empirical reference tunes, AR23 and the MicroBooNE tune (drawn dashed in the projected panels). The samples are summarised in Table 1.

Panels (a)–(d) use the CC $1\mu 1p$ transverse-kinematic-imbalance (TKI) sample: the single-transverse momentum imbalance δp_T and boosting angle $\delta\alpha_T$, with the two-dimensional δp_T shown in the most back-to-back ($\delta\alpha_T < 45^\circ$) and most FSI-disturbed ($135^\circ < \delta\alpha_T < 180^\circ$) angular slices. Panels (e)–(f) use the CC $1\mu Np$ sample: the muon momentum p_μ and the leading-proton momentum p_p . The TKI observables isolate nuclear effects: δp_T is driven by Fermi motion and FSI, while the migration of strength toward $\delta\alpha_T \rightarrow 180^\circ$ reflects the FSI-induced momentum loss of the outgoing proton.

Replacing the empirical references with the explicit N/LFG ground state, lattice F_A^{LQCD} , and hA2018 cascade, `LFG26_24a_00_000` reproduces the bulk of the δp_T distribution and the p_μ/p_p spectra. At the δp_T peak it sits highest of the three predictions and closest to the data, whereas AR23 underpredicts the peak; the predictions converge in the high- δp_T tail where single-nucleon kinematics no longer dominate.

GENIE sample	Nuclear model	Role
<code>LFG26_24a_00_000</code>	$\text{N/LFG} + F_A^{\text{LQCD}} + \text{hA2018}$	this work (solid)
<code>AR23_20i_00_001</code>	AR23 tune	reference (dashed)
<code>G18_10a_02_12a</code>	MicroBooNE tune	reference (dashed)

Table 1: The three GENIE samples overlaid on the MicroBooNE data. The *a/b* suffix in the `LFG26` naming selects the FSI model (here **a** = hA2018); the leading digits select the ground state (LFG) and the axial form factor (F_A^{LQCD}).

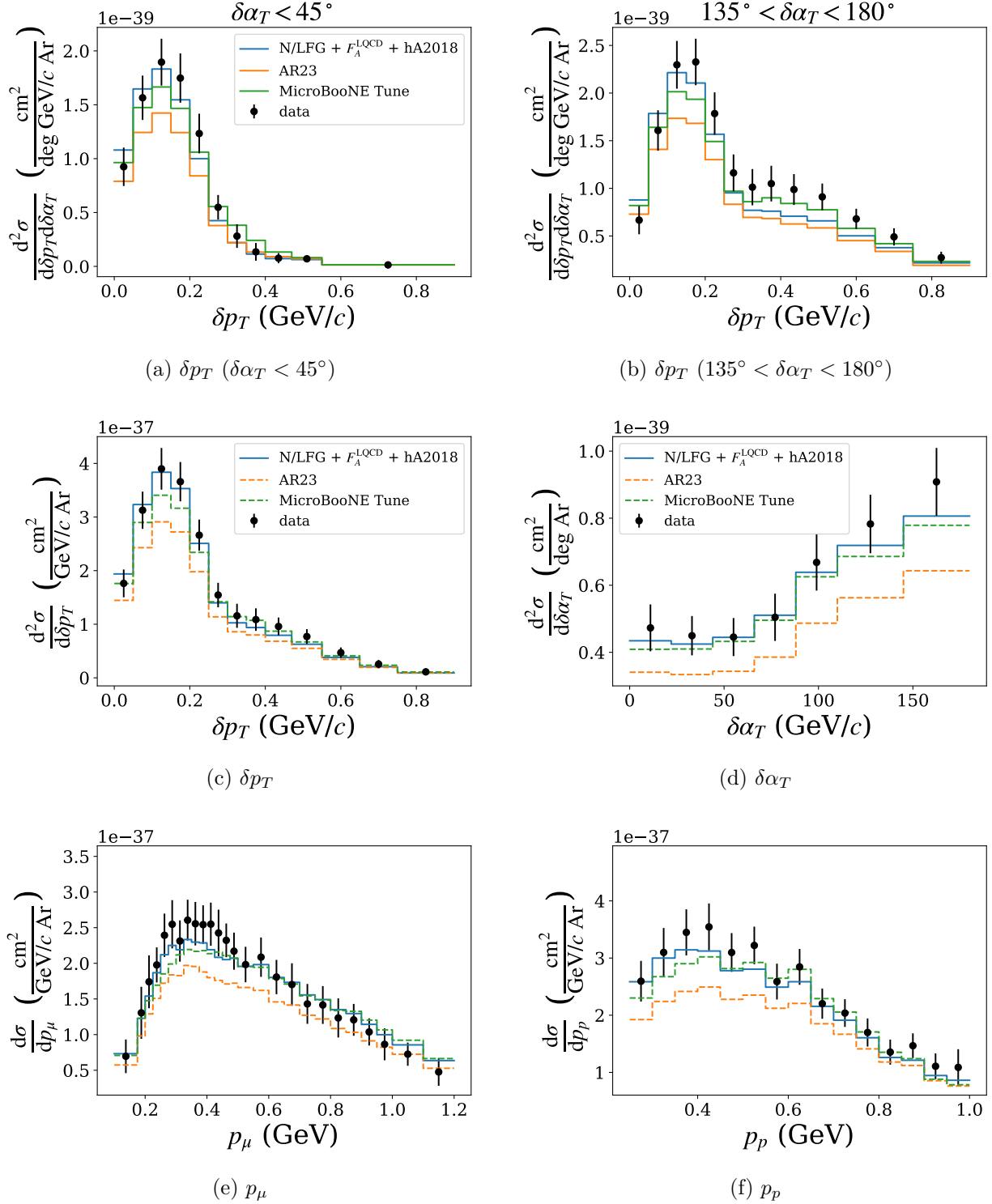


Figure 1: MicroBooNE charged-current cross sections (black points) compared with the N/LFG + F_A^{LQCD} + hA2018 model LFG26_24a_00_000 (solid blue) and the AR23 and MicroBooNE reference tunes (dashed in panels (c)–(f)). (a)–(b) two-dimensional δp_T in the lowest and highest $\delta\alpha_T$ slices of the CC $1\mu 1p$ TKI measurement; (c)–(d) the projected δp_T and $\delta\alpha_T$; (e)–(f) the muon and leading-proton momenta from the CC $1\mu Np$ measurement.